TIORA

The Existence of Time Depends on the Stability of the Cosmic Environment

A Speculative Model of Time Rate, Spacetime Tearing, Black-Hole Information Output, Relative Time Acceleration, and One-Way Travel to the Future

$$q = d\tau/dt \qquad D \geq D_c$$

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Theory Note

This paper advances a core conjecture: the existence and rate of time may depend on the stable state of the cosmic environment. I refer to this not-yet-defined environment as tiora.

From this premise, the paper develops a series of conjectures concerning gravity, black holes, white holes, spacetime tearing, information output, high-energy events, and time engineering, and examines whether these phenomena can be connected by a common underlying mechanism. It also considers a deeper possibility: whether the experimentally verified relations of special relativity (SR) and general relativity (GR) concerning motion, proper time, gravity, and spacetime geometry might be understood as macroscopic manifestations of tiora under different stable states and disturbance scales.

At present, tiora has no experimental detection, complete mathematical definition, or independent field equation. It is therefore presented not as a replacement conclusion for existing relativity, but as an underlying conjectural framework awaiting derivation, testing, and falsification. References to relativity, black-hole observations, dark matter, dark energy, and other current research are used only to establish constraints, comparisons, and possible tests.

Abstract

This paper introduces a speculative cosmic environmental variable provisionally named **tiora**. Its microscopic ontology is left open: it may be represented as a background field permeating the universe, an environmental component with an effective "concentration," or a deeper rule of spacetime. The central assumption is that the stable existence and progression of time require local tiora to remain within a stable range. The stable state of the ordinary universe is denoted by T_0, with the relative time rate defined as $q = d\tau/dt = 1$. Local temporal structure may become unstable when tiora lies either above or below this stable range. When the deviation exceeds a critical threshold and q undergoes a nonlinear departure from 1, the resulting severe desynchronization of causal output between the local region and the external universe is defined here as a **spacetime tear**. The black-hole analysis adopts a slow-time working branch in which q falls sharply, while retaining a fast-time branch in which q may increase. The slow-time branch can produce extreme redshift, very low photon arrival rates, long apparent lifetimes, and causal-output bottlenecks. The framework also yields several testable conjectures: small-scale high-energy events may produce transient tears; a very short local history inside a black-hole-like region may map to an extremely long external observational time; matter, energy, and information may continue to evolve as long as $q > 0$; and if tiora can be engineered, local time slowing, relative time acceleration, and one-way travel to the future may become engineering consequences of the model.

Keywords: tiora; time rate; spacetime tear; black hole; information output; temporal layering; dark energy; event horizon; future travel

1. Research Starting Point and Core Hypotheses

This paper defines the scope of the tiora model through five core hypotheses and uses them to organize the later definitions, black-hole model, and falsifiable predictions.

1. The local physical history of a black hole may be far shorter than the apparent lifetime inferred by a distant observer. Milliseconds, seconds, or minutes of local evolution could correspond to billions of years or more of external time under an extreme time-rate mapping.
2. Black-hole darkness may arise from an extreme stretching of the causal-output timeline: light, energy, and geometric state may continue to evolve locally while the amount of new information received per unit external time approaches zero.
3. Small-scale high-energy events may also cross the critical tiora threshold briefly. If the affected region is tiny, the deviation in q is limited, and recovery is rapid, such transient tears may never have been identified as black-hole-like phenomena.
4. If tiora can be manipulated and a stable low-q region can be engineered, one-way travel to the future may follow from a sustained difference between local proper time and external reference time.
5. The direction of the time-rate change after tiora instability is not yet fixed. Both $q < 1$ and $q > 1$ are retained as possible branches, with their domains to be decided by observation and future coupling equations.

These hypotheses share one starting point: time may depend on a cosmic environment that can vary locally. The value q = 1 in the ordinary universe is used only as a comparison baseline and is not assumed to be the maximum or minimum possible time rate.

2. Provisional Ontology of Tiora

The hypothetical cosmic environment is denoted by **tiora**, while its ontology is intentionally left open. Tiora may be represented as a field, an effective environmental component, or a deeper rule. At the present stage, the model only requires these descriptions to support a stable state, local deviations, and a relation to the rate of time.

- **Field interpretation:** tiora is a background field permeating the universe, whose local state affects how quickly time can progress.
- **Matter/concentration interpretation:** tiora may be treated as an environmental component with an effective "amount" or "concentration." Here concentration is only a macroscopic variable and does not presuppose tiora particles.
- **Rule/law interpretation:** tiora may be understood as a deeper environmental condition that allows local spacetime to operate according to stable rules, without requiring an ordinary material microstructure.

At this stage the model focuses on two macroscopic questions: how the tiora state determines the local time rate, and how high-energy events alter that state and trigger subsequent recovery.

3. Four Basic Quantities: Tiora State, Deviation, Environmental Integrity, and Time Rate

The model treats **environmental integrity** and **time rate** as distinct variables. Environmental integrity describes how far tiora departs from its stable range, whereas q describes how quickly local proper time accumulates relative to the reference time of the ordinary universe. A decrease in environmental integrity therefore does not predetermine whether q must be below 1.

T_0 = stable tiora state of the ordinary cosmic environment (interpretable as a stable value or the center of a stable interval)

$\delta(x,t) = [T(x,t) - T_0] / T_0$ = signed deviation of tiora from its normal value; $D = |\delta|$ denotes the magnitude of deviation/instability

$q(x,t) = d\tau/dt$ = local time rate / reference time rate of the ordinary universe

Here t is the cosmic reference time in a normal tiora environment, and τ is the proper time accumulated by a local observer. The ordinary universe is defined by q = 1. One may also define an environmental-integrity or "time-level" variable L: the normal stable universe has the highest L, and L decreases as tiora moves away from the stable range. A lower L, however, does not imply that q must decrease.

The temporal effect of tiora is controlled by the deviation magnitude D rather than simply by whether the absolute tiora value is high or low. Tiora that is either too low or too high may leave the

stable range. When D is small, q remains close to 1; only after D crosses the critical threshold D_c does q enter a strongly nonlinear regime.

$$D \ll D_c \Rightarrow q \approx 1$$

$$D \geq D_c \Rightarrow |q - 1| \text{ increases rapidly (direction not yet determined by the model)}$$

Slow-time working branch: as D increases, $q \to 0$ while remaining $q > 0$.

Fast-time retained branch: increasing D may instead produce $q > 1$ with a rapid rise.

The choice between these branches must ultimately be fixed by coupling equations and observations rather than intuition.

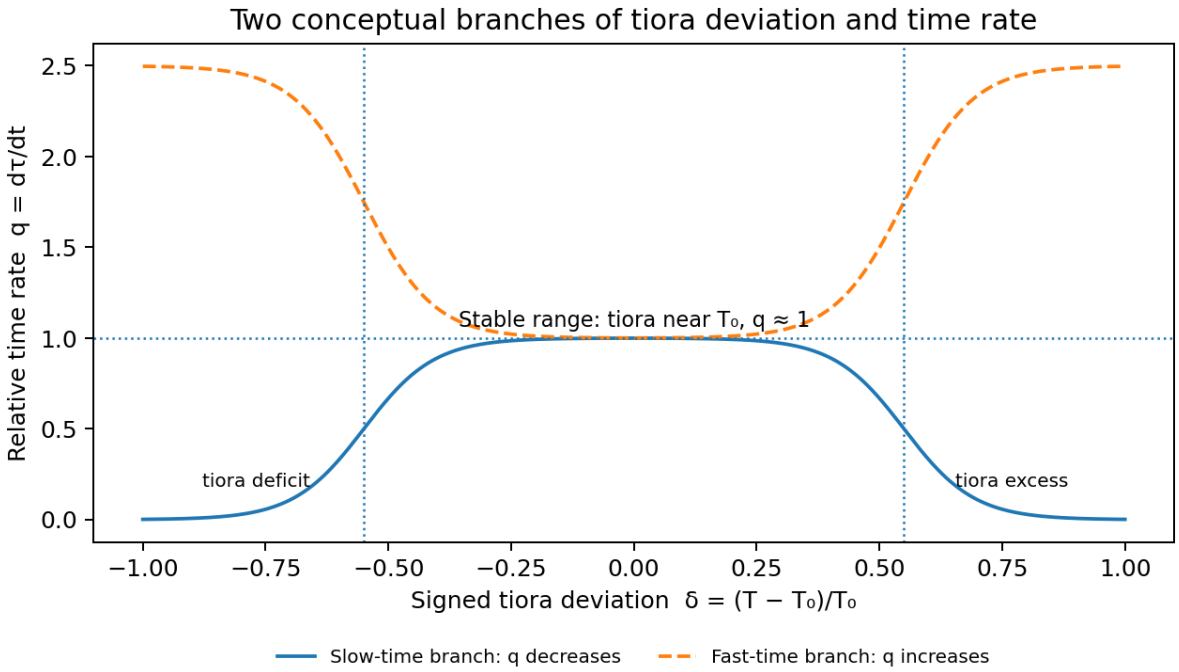


Figure 1. Two conceptual branches of tiora deviation and time rate. Either an excess or a deficit of tiora may cross the stable threshold. The q-decreasing branch is used for the black-hole model, while the q-increasing branch is retained as an open possibility. The curves are schematic, not experimental fits.

4. Operational Definition of a "Spacetime Tear"

The term **spacetime tear** is used here to describe a critical state in which time rate and causal output undergo a qualitative change. It does not mean that space literally tears like a material sheet.

Spacetime tear: a state in which local tiora leaves the normal stable range and exceeds the critical deviation D_c, causing the relative time rate q to depart nonlinearly from q = 1 and producing severe desynchronization between the causal-output timeline of the local region and that of the external universe. In the slow-time branch, q falls sharply and the outward updates of light, matter, energy, and geometric state are stretched to extreme durations. In the fast-time branch, the opposite kind of temporal compression may occur. The black-hole model developed here primarily adopts the slow-time branch.

D_c marks the onset of the strongly nonlinear regime; it does not presuppose one-way depletion of tiora. Either an excess or a deficit of tiora may cross the threshold. D_c is also not equivalent to $q = 0$. The limits $q = 0$ and $q \to \infty$ are retained only as mathematical limits and are not treated as finite physical states without a complete equation set and experimental support.

5. Energy-Limit Hypothesis in the Slow-Time Branch

The slow-time working branch adopts a conditional assumption: as tiora instability approaches an extreme state, the energy required to increase D further rises very rapidly. The growth may be exponential or steeper; no specific function is imposed at this stage.

E(D) grows superlinearly with D, with the possible limiting behavior $\lim(D \to 1) E(D) = \infty$.

For example, increasing the deviation from a moderate level to a near-extreme level may require energy changes spanning many orders of magnitude. Any percentages used in illustrative examples represent only nonlinearity and are not measurements.

Under the slow-time branch, finite energy may drive q arbitrarily close to 0 without requiring q to equal 0. Local time therefore remains defined, and matter, energy, and information continue to evolve in proper time τ. If future equations instead favor a q-increasing branch, the detailed limiting function in this section would need revision, while the broader modeling principle—that finite energy need not reach a mathematical singularity—may still be retained.

6. Local Proper Time and Cross-Region Comparison

If an observer's thought processes, biological processes, atomic transitions, ideal clocks, and optical processes all share the same proper time τ, then local time feels normal. Differences in q become measurable only through comparisons between different regions or different worldlines.

This structure resembles locality in general relativity: a freely falling observer in a sufficiently small laboratory does not directly experience an overall rescaling of their own local time rate.[1] Tiora additionally requires a rule for mapping information across regions: after an external signal enters a region with a different q, is its arrival rate and playback rate retimed by the receiving environment? A low-q observer might therefore see incoming information at an approximately normal playback rate, or might instead see it accelerated. The result must be derived from a propagation law.

Local: $c_{local} = c$; a local clock records τ normally.

Cross-region comparison: $q_1 \neq q_2$.

Regions in different tiora states may accumulate different amounts of proper time τ, while the arrival rate of cross-boundary signals may also depend on tiora gradients and propagation rules. This issue determines the detailed outcome of thought experiments such as a low-q region observing the external universe or the external universe observing a black-hole interior.

7. Minimal Geometric Expression: From Time-Rate Gradients to Changes in Spatial Paths

The causal ordering adopted in this paper is: the tiora state changes the local rate at which time progresses; different values of q form a spatial gradient; the motion of matter and light then departs from the uniform structure of the ordinary universe; on macroscopic scales, this appears as effective spatial geometry and gravitational behavior.

As a purely illustrative expression, consider the simplified spherically symmetric metric

$ds^2 = -q(r,t)^2\, c^2dt^2 + B(r,t)^2dr^2 + r^2d\Omega^2$.

This metric is not derived from tiora dynamics. It only makes the intended structure explicit: q controls the temporal component, while B represents the geometric response in the spatial direction. A complete theory must determine q and B from the same tiora state and provide their coupling relation.

For a local light ray, $ds^2 = 0$. In this toy expression, the radial propagation rate measured in distant coordinate time contains roughly a q/B factor. When q is extremely small, a distant observer may see an extremely small information-output rate even though a local observer still measures the speed of light as c.

8. The Coupling Problem Between Mass-Energy and Tiora

General relativity relates energy density, momentum, pressure, stress, and related quantities to geometric curvature through Einstein's field equations.[1] If tiora is introduced as a more fundamental variable, a complete model must specify how stress-energy changes tiora and how tiora in turn determines time rate and geometry.

Three candidate structures are retained:

- **Coupling model:** mass-energy directly changes the local stable state of tiora, and tiora then determines time rate and geometry.
- **Storage/deformation model:** some portion of what is called "mass" may correspond to a stable tiora deformation or field-energy state; denser matter would then produce a stronger local change in the tiora environment.
- **Common-source model:** mass and tiora may both be macroscopic manifestations of a deeper structure, so they are related without a simple one-way production relation.

No explicit coupling equation between mass-energy and tiora has yet been established. This is one of the central theoretical gaps of the current model. Without such an equation, the model cannot independently compute q, light paths, orbits, or black-hole scales from a given stellar mass.

9. Working Hypothesis: Tiora Consumption/Coupling in Stars

The current version adopts the working hypothesis that stars continuously consume, absorb, or cycle tiora during their lifetimes. "Consumption" here means that tiora enters some internal storage, coupling, or conversion state within the star; it does not assume permanent destruction.

During stable stellar burning, tiora exchange may remain relatively smooth. Stellar death, collapse, explosion, or a violent phase transition may trigger the rapid release, redistribution, or local shortage of tiora that had previously been stored or involved in the cycle. The triggering condition for a black-hole-like state is whether the local deviation leaves the stable range within a sufficiently short time.

Stable exchange → small deviation; sudden release / shortage / redistribution → rapid growth of $|\delta|$ → possible crossing of D_c from either direction.

Stellar death does not necessarily produce a black-hole-like state. A tiora disturbance has parameters such as depth, volume, and rate of change; only when the local state enters the nonlinear regime $D \geq D_c$ does the model predict a macroscopic black-hole-like phenomenon.

10. Small-Scale Transient Tears

Because a spacetime tear is defined by tiora deviation crossing D_c and triggering a nonlinear change in q, the logical condition is not restricted to stellar scales. A very small region with sufficiently high energy density or a sufficiently rapid tiora disturbance could also cross D_c for a very short time.

Such transient tears may be characterized by a tiny spatial scale, a limited deviation of q, and very rapid recovery. The slow-time branch could produce external delays on nanosecond, microsecond, or shorter scales; the fast-time branch could produce equally brief temporal compression. Their observational appearance could differ substantially from long-lived astronomical black holes.

This paper distinguishes **astronomical black-hole candidates** from **black-hole-like states in the tiora model**. The former are long-observed compact strong-gravity objects; the latter merely indicate that a local region has entered the $D \geq D_c$ spacetime-tear regime, which may last for an extremely short time.

11. Slow-Time Black-Hole Branch: Unstable Region and Outside-In Recovery

The astronomical black hole is represented here by a slow-time tiora-tear working branch: the local tiora environment is severely destabilized, q is extremely small near the center, and q gradually approaches 1 with increasing radius. This branch is used to test whether extreme redshift, darkness, long apparent lifetime, and a causal-output bottleneck can arise from one temporal structure; other directions of tiora instability remain open.

In a spherically symmetric illustration, the central region has the deepest tiora deviation, which weakens outward and eventually transitions smoothly to the ordinary universe. The slow-time branch therefore produces a radial $q(r)$ gradient: q is extremely small near the center and gradually approaches 1 toward the exterior.

The normal tiora environment then restores the unstable region from the outside inward. The outer layers are less displaced and have q closer to 1, so they return to the stable range earlier. The

inner region has deeper deviation and lower q, making its recovery extremely slow when described in external reference time.

R_tear(t): radius of the region satisfying D(r,t) ≥ D_c.

During recovery: R_tear(t) gradually decreases.

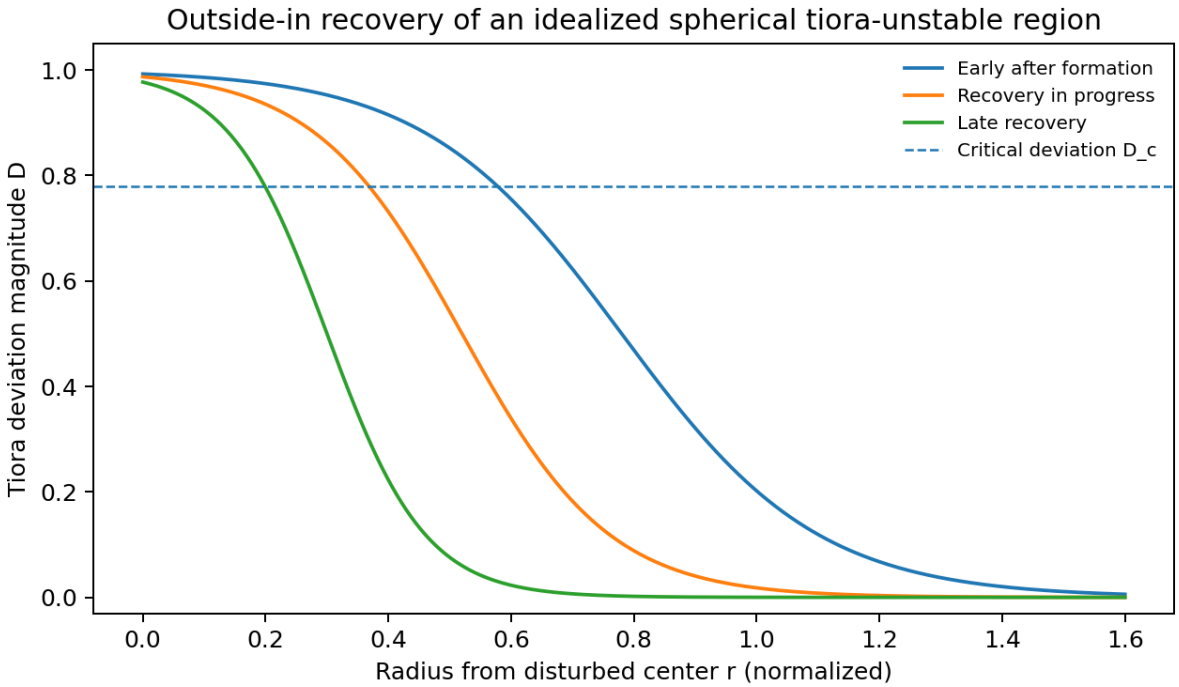


Figure 2. Outside-in recovery of an idealized spherically symmetric tiora-unstable region. D denotes the magnitude of deviation from normal tiora and may arise from either excess or deficit. Only the slow-time black-hole branch is shown; the radius above threshold shrinks during recovery.

12. Continuous Tiora Gradient Around a Black Hole

D_c marks only the onset of strongly nonlinear temporal effects. Weaker tiora deviations may still exist outside the threshold region, forming a broader transition zone and participating in the recovery of the interior unstable region.

In this transition zone q may differ from 1 by only a tiny amount, for example 0.999999… . Such a magnitude may be indistinguishable from standard gravitational effects with present observations. Larger percentages used in early conceptual examples are only mechanism illustrations; the real amplitude must be observationally constrained.

This yields a continuous radial structure: a strongly torn central region, a more weakly unstable outer zone, and a gradual return to the ordinary universe farther out. Because information from different radii experiences different delays, a single black-hole image may contain signals from different local historical times, producing **radial temporal layering**.

13. Black-Hole Darkening: Stretching the Causal-Output Timeline

The slow-time branch keeps q strictly greater than zero, so light, matter, energy, and information can continue to evolve in local proper time. In this model, black-hole darkness arises primarily from the stretching of the causal-output timeline: the amount of new information received per unit external time approaches zero.

$q = d\tau/dt \Rightarrow dt = d\tau/q$.

If the unstable region experiences only 3 seconds of local proper time while its average q is extremely small, that local history may map to an enormous interval of external time. A black hole observed for a very long time therefore need not have undergone an equally long local physical history.

The mechanism can be summarized as follows:

An extremely short local black-hole history may map to a cosmic-scale external observational history under sufficiently low q.

If a local luminous process emits N_{local} detectable photons per local second, the external arrival rate may be written conceptually as $\dot{N}_{obs} \approx q_{eff} \cdot \dot{N}_{local}$, with additional factors for redshift, geometric dilution, and propagation-path effects. Under extremely small q_{eff}, a continuous local photon stream can be stretched into an extremely low external arrival rate, producing an observationally dark region. All numerical examples in this paper are illustrative.

This also gives the threshold a direct observational meaning. Outside the threshold, q remains close to 1 and surrounding matter may continue to appear normally dynamic. After entering the nonlinear low-q region, the same short segment of optical history can be stretched across astronomical durations, producing a visually sharp dark region.

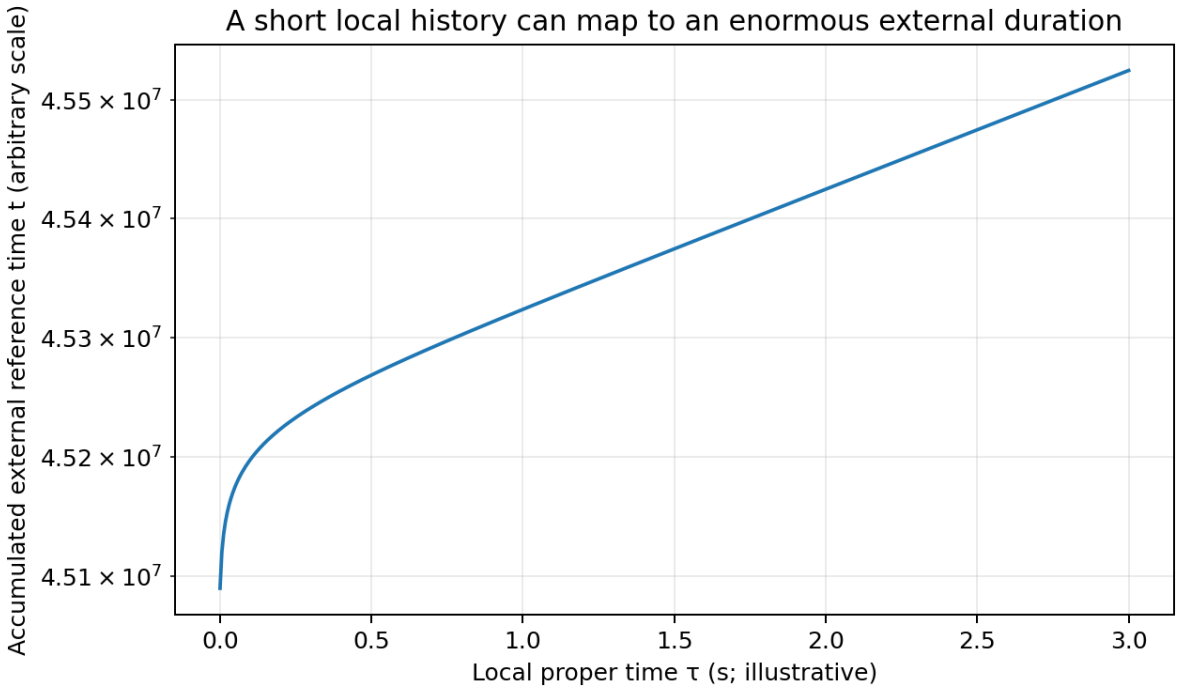


Figure 3. Illustrative mapping from a short local history to a very long external duration in the slow-time branch. Values show only the mathematical possibility of extreme temporal stretching and are not estimates of real black-hole lifetimes.

14. Consistent Delay Across All Causal-Output Channels

Internal consistency requires different causal channels to obey the same local q structure. If optical information still describes an earlier state while geometric changes reach the exterior without delay, a distant observer would receive mutually inconsistent descriptions of the system.

Accordingly, light, electromagnetic signals, matter outflow, energy transfer, the remote influence of geometric changes, and changes in the tiora state itself must all be mapped to external reference time by a unified propagation law.

$$\Gamma_{out,external} \propto q \cdot \Gamma_{out,local} \quad \text{(conceptual relation only)}$$

Under this condition, the unstable region may already have completed its recovery in local proper time while a distant observer continues to receive causal information from much earlier stages. The delay applies to the entire state update, not only to visible light.

15. Information Mapping Between Regions with Different q

Cross-region observation requires both the local time rate and the propagation of signals to be described. Let observer A occupy a region with time rate q_A, and let the observed region B have q_B. If boundary and propagation effects are temporarily ignored, the relative playback rate is first expected to depend on q_B/q_A.

This paper additionally introduces a working hypothesis of **input synchronization–output dilation**: after a signal enters a new tiora environment, its local arrival and playback rhythm may be retimed by the receiving region. Matter, ideal clocks, observers, and incoming information inside a low-q region may therefore share the same low-q rhythm.

For example, suppose Earth is artificially adjusted to $q_E = 0.01$ while the external universe remains $q_U = 1$. Using q_U/q_E alone would suggest a dramatic fast-forward view of the universe. If cross-boundary signals are synchronously remapped upon entering the low-q Earth region, then the observer's thoughts, clocks, and incoming information all progress according to the local rhythm, and the subjective playback rate may remain close to normal. Absolute time labels, propagation distance, and historical delay between the regions may still differ.

For output from the low-q Earth to the $q_U = 1$ exterior, the signal is generated and exits from within a low-q environment, so its output timeline may already be stretched at the source. The external observer would then receive a strongly slowed history of Earth. This directional rule is summarized here as **low-q input synchronization–low-q output dilation**.

The same rule can be applied to the slow-time black-hole branch. If the interior satisfies $q \ll 1$, incoming external information may be retimed by the local environment, so an interior observer need not see the universe moving infinitely fast. Information generated inside the low-q region, however, is output at that low local rate and therefore appears extremely slowed to a distant observer at $q = 1$.

To preserve room for a mathematical formulation, the cross-boundary propagation factor $P_{B\to A}$ is introduced, giving the conceptual relation $R_{B\to A} \sim (q_B/q_A) \cdot P_{B\to A}$. P is an undefined propagation variable representing how signals are remapped when crossing a tiora boundary. A complete theory must determine P from a unified propagation equation and test the input-synchronization hypothesis experimentally.

Because this hypothesis may introduce a stronger background structure than standard relativity, it must satisfy existing limits on local Lorentz invariance under ordinary conditions $D \ll D_c$ and must treat all propagation directions with one consistent rule.

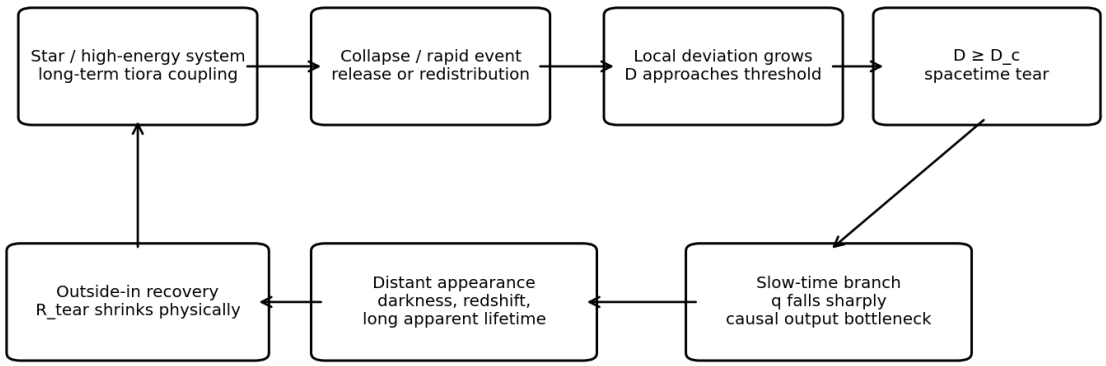
16. Information Catch-Up: Output Compression After Rapid Recovery

Consider an artificially low-q region that later returns rapidly to q = 1. The treatment of light, matter disturbances, and other causal updates that were previously stretched in time defines the core problem of the **information catch-up effect**.

One candidate mechanism is temporal catch-up: if earlier wavefronts or state updates remain stored within the low-q output channel rather than being destroyed, rapid recovery of q may cause the receiver to obtain a dense sequence of a much longer local history within a shorter interval, appearing as temporal compression or rapid playback.

This mechanism requires an explicit carrier and propagation model for information and must calculate possible energy-flux peaks, additional redshift or blueshift, and causal consistency. If no physical storage or propagation structure exists, the system may simply begin to output normally from its new state after recovery without replaying all older information. These branches must ultimately be distinguished by the dynamics.

Formation and recovery logic of the current slow-time black-hole branch



Recovery proceeds physically, while outgoing state updates remain constrained by the local q structure.

Figure 4. Formation and recovery logic of the current slow-time tiora black-hole branch. The black-hole-like state is an unstable environment; darkness arises from a causal-output bottleneck associated with extremely low q.

16.1 White-Hole-Like States: A Tiora Mechanism Corresponding to the GR White Hole

The term **white hole** in this section follows its general-relativistic meaning. In standard GR, a white hole can be described as the time-reversed structure of certain black-hole spacetimes, in which matter and signals may emerge outward but cannot ordinarily enter from the exterior. Historical quantum-field analyses indicate that white-hole backgrounds may involve extreme particle and energy fluxes and significant stability problems.[21] Tiora introduces a different possible underlying mechanism while leaving the GR geometric definition unchanged.

In the slow-time black-hole branch, $q \ll 1$ stretches a finite local history into extremely slow external output. The corresponding tiora white-hole conjecture uses the reverse form of temporal compression: a region might enter $q \gg 1$, or information accumulated during a low-q phase might undergo extreme information catch-up during recovery, compressing light, matter, energy, and geometric updates from a long history into a very short external interval. The two theoretical routes differ, but both may generate intense outward output and severe difficulty maintaining stability.

Within the tiora framework, the stability of a white-hole-like state may be related to the time-rate multiplier. Low q suppresses the rate at which local changes are unfolded into external time, allowing a black-hole-like state to appear long-lived. High q or extreme temporal compression increases the amount of state change, energy release, and feedback per unit external time, potentially producing high flux and strong backreaction. This is a tiora-specific stability conjecture independent of standard analyses of white-hole stability.

A further observational conjecture is retained: if electromagnetic information from many different times, wavelengths, and colors is compressed into an extremely short output interval, the receiver may encounter an enormous broadband photon flux. Given the finite temporal and color integration of human vision, such an event could appear as an extremely bright, near-white flash. The historical term “white hole” is not based on visible color; this visual prediction belongs only to the tiora extension.

17. Historical Information Layering and the Current Material State

The simultaneous arrival of historical optical information does not imply duplication of matter. At any given local moment, the physical object has only one current state, while distant observation depends on signals emitted at earlier times. Different propagation paths and different amounts of temporal stretching can cause multiple historical stages of the same object to appear within one observational window.

Real astronomy already contains an analogous observational structure: Hubble has recorded multiple historical stages of the same supernova in a single exposure because gravitational lensing created different propagation paths and time delays.[8] This fact is not evidence for tiora, but it shows that multiple historical optical states of one object can reach an observer at the same time without violating established physics.

Tiora further conjectures that an extreme q gradient may produce stronger historical layering. The exterior may display more recent states, deeper regions may display older states, and the deepest region may provide almost no new light. The actual matter in a region may already have recovered and moved into a new state while the observer continues to receive information from its earlier black-hole phase.

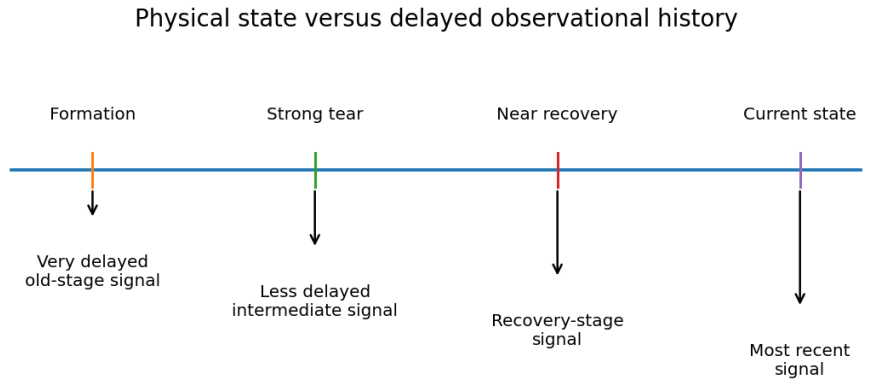


Figure 5. The physical system has only one local “now,” while a distant observer may receive delayed information from several historical stages within a single observing window.

18. Physical Radius and Observed Radius of a Black Hole

If the recovery boundary moves from the outside inward, the physical radius satisfying $D \geq D_c$, denoted R_{tear}, decreases with time. A distant observer estimates black-hole size from light and other causal information that has already arrived, so a separate apparent radius R_{obs} can be defined.

$R_{physical}(t)$ = current physical radius that remains in the strong-tear regime.

$R_{observed}(t)$ = apparent radius inferred from historical information that has reached the exterior.

In the tiora model, $R_{physical}$ and $R_{observed}$ may differ. $R_{physical}$ may already be small while $R_{observed}$ continues for a long time to display an earlier black-hole scale.

The difference between $R_{physical}$ and $R_{observed}$ provides a candidate mechanism for black-hole apparent stability, but it must be calculated explicitly. The model must provide the evolution $R_{observed}(t)$ and compare it with EHT and related observations rather than using “observational delay” as a substitute for quantitative prediction.

19. Continued Existence of Matter, Energy, and Information Inside the Black Hole

The current version adopts a slow-time branch with $q > 0$ rather than setting local time exactly to zero. Matter, energy, and information entering the strong-tear region therefore remain defined and continue to evolve along local proper time τ.

Finite energy in this branch only drives q toward zero without reaching q = 0. Matter and energy inside the strong-tear region continue to exist, and information continues to evolve through local physical processes. What changes is the reference-time scale required for subsequent state updates to propagate into the external universe.

In this branch, the black-hole information problem is reformulated as a problem of the **causal-output timescale**:

Seconds, milliseconds, or even shorter intervals of local physical history may be stretched into long-term external output by extremely low q, while the information remains present but enters the distant observer's causal reach at a drastically reduced rate.

The relevant quantity is therefore the time required for information to re-enter the causal reach of a distant observer, rather than whether it continues to exist during local evolution.

20. Relation to General Relativity: Hypothesis of an Underlying Environmental Mechanism

General relativity has passed many high-precision tests in the Solar System, binary systems, gravitational waves, and strong-gravity observations. The LIGO–Virgo–KAGRA measurement of the GW250114 black-hole ringdown is consistent, at current precision, with Kerr black holes and GR.[6] EHT measurements of the ring scale of Sgr A* are also consistent with Kerr predictions at current precision and constrain some alternative models with material surfaces.[4]

The intended position of tiora is upstream of these verified macroscopic relations. Einstein's equations specify the relation between stress-energy and geometry; tiora hypothesizes that this relation may correspond to a deeper environmental degree of freedom or rule.

A complete tiora theory must therefore reproduce all verified GR results in the regimes where they have been tested, while producing distinguishable predictions only in regions that have not been directly probed or remain theoretically open.

20.1 Required Limiting Structure for SR and GR

Special relativity describes inertial motion and local causal structure in flat spacetime. General relativity extends this structure to dynamical curved spacetime. In a sufficiently small freely falling region, GR locally reduces to the structure of SR, a central implication of the equivalence principle.[1]

Tiora attempts to describe these two limits through one underlying environment. When $T \approx T_0$ and $\nabla T \approx 0$, the model must produce approximately Minkowskian local spacetime and the relations of SR. When mass-energy produces a slowly varying spatial profile T(x,t), q(x,t) and spatial geometry must change accordingly and recover, at macroscopic scales, the curvature, free fall, and gravitational time effects described by GR.

The hierarchy can be summarized as follows:

Uniform stable tiora → SR-like local spacetime

Mild tiora gradient with $D \ll D_c \rightarrow$ GR-like gravity

$D \geq D_c \rightarrow$ strongly nonlinear spacetime tear proposed by the tiora model

20.2 Weak-Field Gravity and Orbits: Compatibility Expressions in the q Language

If tiora is to describe ordinary gravity, its weak-field limit must reproduce Newtonian gravity and verified GR results. In a static weak field one may write $q(r) \approx 1 + \Phi(r)/c^2$, where Φ is the Newtonian gravitational potential. Near Earth or the Sun, $\Phi < 0$, so q is lower near the mass source and approaches 1 at large distance. This relation belongs to the established GR/Newtonian weak-field structure and serves as a consistency condition that a future tiora equation must reproduce.

Define $\Phi_T \equiv c^2 \ln q$. When $|\Phi|/c^2 \ll 1$, $\Phi_T \approx \Phi$. The Newtonian Poisson equation can then be rewritten as

$$\nabla^2(c^2 \ln q) \approx 4\pi G\rho,$$

and the free-fall acceleration as

$a \approx -c^2 \nabla \ln q$.

These relations are primarily reparameterizations of known weak-field results. They test the compatibility of the q language with standard gravity but do not constitute independent tiora dynamics.

The independent theoretical content of tiora must come from equations further upstream: how mass, energy, pressure, momentum, and energy conversion modify T; how T and D determine q; and how q and spatial geometry jointly determine worldlines. If those equations can derive $q \approx 1 + \Phi/c^2$, the inverse-square law, gravitational redshift, and orbital behavior without importing the GR result in advance, then tiora would acquire genuine explanatory content beneath those macroscopic relations.

20.3 Airborne Atomic Clocks and Energy Conversion: A Weak-Disturbance Tiora Hypothesis

The Hafele–Keating airborne atomic-clock experiment and later higher-precision clock experiments agree with the time differences predicted by SR and GR.[17] Within the tiora framework, these relativistic relations are treated as macroscopic laws that must emerge in the weak-disturbance limit. The additional conjecture made here is that the energy input, conversion of energy forms, and change in the energy-momentum state during a change of motion may alter the local tiora-coupled state and thereby establish the corresponding q.

This hypothesis does not treat the numerical value of velocity itself as an independent tiora source. When a body changes from one state of motion to another, chemical, electromagnetic, or other forms of energy may be converted into the system's kinetic energy while the system's energy-momentum state changes. The paper conjectures that this process couples to tiora. For an ordinary aircraft, the energy scale is small, corresponding to $D \ll D_c$ and $q \approx 1$, so only an extremely weak difference in time rate would result.

The mechanism must also explain the persistence of kinematic time dilation after acceleration ends. Once acceleration is complete, the system no longer receives the same continuous power input,

but its energy-momentum state has changed. The tiora model therefore requires a **state-coupling** relation rather than a **continuous-power** relation and must derive from its own equations the Lorentz factor, frame symmetry, and the proper-time relation between worldlines. If it cannot do so, this hypothesis would amount only to a relabeling of SR results.

The present working chain is

energy conversion / energy-momentum state change → change in tiora-coupled state → change in q → proper-time difference.

If a complete equation set can reproduce SR exactly at ordinary energy scales while naturally driving D toward D_c at very high energy density or during sufficiently violent energy conversion, airborne clock experiments, high-energy particles, stellar collapse, and spacetime tearing could be placed on one continuous scale of disturbance strength.

21. Event Horizon and Extreme Temporal Bottleneck: Two Causal Structures

In standard GR, the event horizon of a black hole is a strict causal boundary: once an event lies inside the horizon, no future-directed light path reaches external infinity. NASA's standard description refers to it as a "point of no return" from which even light cannot escape.[2]

If the current tiora slow-time branch always satisfies $q > 0$, it more naturally produces a finite but extremely long causal-output timescale rather than a permanently absolute causal boundary. Within practical observing times, such a region could still appear almost completely unable to output new information.

This structure has a formal resemblance to theoretical **black-hole mimickers**. Research published in Physical Review D in 2026 continues to examine horizonless spacetimes capable of reproducing black-hole shadows and ringdowns.[7] Tiora would still need to provide its own distinguishable signals.

EHT observations of Sgr A* already exclude some simple thermal-surface models and strongly constrain perfectly reflecting surfaces.[4] A tiora unstable region therefore cannot behave as an ordinary material surface; its exterior geometry and radiation must remain close to Kerr within current observational precision.

21.1 Echoes: Testing Very-Late-Time Signals

If no strictly permanent event horizon exists and q remains positive, some disturbances may re-enter the causal reach of distant observers after very long delays. This structure can produce candidate signals related to gravitational-wave echoes. Research in 2026 continues to calculate long-lived modes and echo structures for horizonless ultracompact objects.[19]

An echo would not be unique to tiora; other horizonless models and some treatments within GR can also produce echoes.[20] Discriminating power would therefore come from predictions made in advance for delay, frequency drift, and damping, and from whether the same q(r,t), R_tear(t), and recovery function $\mathscr{R}$(D) can explain all of these quantities once parameters are fixed.

For example, if black-hole candidates of different masses and formation histories obey one scalable relation such as

$$t_echo = F[q(r,t), R_tear(t), \mathcal{R}(D)],$$

and this relation is specified by the tiora equations before the observations are analyzed, then echoes could become a model-discriminating test.

21.2 The Light-Speed Barrier and D_c: An Open High-Energy Coupling Branch

Standard special relativity treats c as the local causal limit. As a massive object approaches c, the energy required for further acceleration grows rapidly, and ordinary continuous acceleration cannot bring it to or through c.[22] Tiora retains an underlying coupling conjecture at this point: the extreme energy-momentum state accumulated near this limit may also drive a growing local tiora deviation.

In this deliberately bold branch, continued acceleration raises the system's energy-momentum scale, while the mass-energy/tiora coupling increases D. D might approach D_c before the conventional light-speed boundary is reached, causing q to enter a nonlinear regime. The model does not currently identify v = c with D = D_c; any relation between the two must be established independently by equations and experiment.

If an effective superluminal phenomenon were ever observed, tiora retains two candidate manifestations. First, after an environmental threshold is crossed, q may drop sharply, making the continuous motion information extremely sparse, dim, or nearly untrackable to a q = 1 observer. Second, the local tiora causal structure may undergo a discontinuous reconfiguration, allowing the object to reappear through a different causal path or state. Neither branch currently has observational support.

In these conjectures, spatial propagation speed and the local rate of proper time belong to different levels of constraint. Even if an effective displacement with v > c were eventually established, local proper time could still be governed by q.

22. Hawking Evaporation and the Information Problem

Classical GR does not make an isolated black hole evaporate automatically. Hawking evaporation arises from quantum field theory in curved spacetime and predicts that black holes possess a temperature, radiate slowly, and lose mass. For known astrophysical black holes, the complete evaporation timescale is vastly longer than the present age of the universe, so the terminal stage has not been directly observed.[3]

Quantum theory requires unitary evolution for a closed system, while the picture of complete black-hole evaporation into thermal radiation raises the question of how information is recovered.[13] A 2026 APS survey of foundational physics likewise reflects the lack of a single settled view of black-hole interiors and the information problem.[14]

The tiora slow-time branch offers a conceptual route: if no strict horizon permanently removes information from the exterior causal domain, and there is instead only an extreme q → 0

delay, information may remain present and be released outward at an extremely low rate. The information problem is then reformulated as a problem of output timescale and consistency with quantum evolution.

This route must still demonstrate compatibility with quantum unitarity, energy conservation, and the known constraints associated with the Hawking thermal spectrum.

23. Structural Correspondence with Quantum-Gravity Models: Short Interior Time and Long Exterior Time

In a 2018 APS Physics discussion of loop-quantum-gravity black-to-white-hole / Planck-star scenarios, Carlo Rovelli noted that extreme relativistic time distortion can allow certain interior processes to last only microseconds while billions of years pass externally.[15] This temporal structure is formally similar to tiora's mapping of a short local history to a very long exterior history.

The mechanisms differ. The cited quantum-gravity scenario relies on a black-hole-to-white-hole transition, whereas tiora relies on the recovery of an environmental variable q and a causal-output bottleneck. The comparison is included only to show that a comparable separation of timescales has already appeared in established theoretical research.

24. Apparent Black-Hole Stability and the Recovery Front

EHT data from M87* in 2017 and 2018 show a broadly stable bright-ring size, and later multi-epoch results likewise indicate a persistent ring scale while brightness and polarization structure can vary.[5] This supports the stability of the Kerr-like exterior geometry over current precision and observing baselines.

For tiora to remain compatible with these observations, changes in R_physical must be mapped by time delay into only very small short-term changes in R_observed, so that observations spanning years to decades continue to show a stable ring scale.

The model must quantitatively calculate the mapping between recovery-front speed and observable image evolution and must be tested against multi-epoch EHT data.

25. Direct Collapse and Non-Stellar Seed Formation Paths

The tiora definition allows any event that produces a sufficiently deep environmental deviation and crosses D_c to create a black-hole-like state. Stellar death is therefore only one candidate trigger.

Recent JWST results show a surface-level structural correspondence with this non-standard-seed possibility. In 2025 NASA reported a supermassive black-hole candidate in the Infinity Galaxy that may have formed through direct collapse of a gas cloud, although the research team did not consider direct collapse definitively established.[11] In 2026 NASA also reported evidence of very massive early black holes existing before their host galaxies had fully developed.[12]

Standard astrophysics already includes heavy-seed and direct-collapse mechanisms. These observations therefore provide only a structural analogy and do not independently support tiora.

26. Candidate Correspondence Between Tiora and Dark Energy

Early versions of the model associated tiora with dark matter, but dark matter clusters strongly in galaxies and large-scale structure, whereas the tiora model requires an environmental variable closer to a pervasive cosmic background. That direct identification was therefore abandoned.

The current version retains the possibility that tiora is related to dark energy or to a deeper structure underlying dark energy. NASA's overview of dark energy continues to treat its physical nature as an open question, with possibilities that include a vacuum-background energy permeating space.[9]

DESI's three-year results in 2025 strengthened indications that dark energy may evolve over cosmic time, although the collaboration has not declared this a definitive discovery.[10] If dark energy is indeed a dynamical background component, its structural character would be closer to a cosmic environment that can deviate and recover.

In 2025 DESI also discussed a hypothesis of cosmologically coupled black holes connecting black-hole interiors, dark energy, and the conversion of ordinary matter.[16] The mechanism differs from tiora, but it shows that black-hole–dark-energy coupling is already an active subject of theoretical research.

The identification of tiora with dark energy therefore remains only a candidate conjecture. If the two are identical, tiora must simultaneously satisfy cosmological dark-energy constraints and local strong-gravity observations, making the proposal more tightly falsifiable.

26.1 Revision of the Dark-Matter Route: Cloud-9

Early versions considered the possibility that tiora was identical to dark matter and that stars might generate or substantially modify the total amount of dark matter. NASA's 2026 Hubble results describe Cloud-9 as a gas-rich, starless, dark-matter-dominated RELHIC candidate.[18] This object motivated the model to abandon the simple route in which tiora equals dark matter and stars are its principal source.

Cloud-9 does not rule out every possible interaction between stars and dark matter, but it shows that standard dark matter need not rely on stars as a universal source in order to dominate a system. A direct identification between tiora and standard dark matter therefore lacks the structural match required by the model.

Tiora instead requires a broadly distributed background variable associated with the normal temporal environment and capable of local deviation and recovery. Dark energy is therefore a closer candidate analogy. DESI's indications of potentially dynamical dark energy provide additional structural motivation, but they are not evidence that the two are identical.[10]

27. Directional Observations of Radial Motion and Time Gradients

If q varies with radius, an observer moving toward a lower-q region and one moving toward a higher-q region should obtain direction-dependent comparisons of incoming light frequency, event-

arrival rates, and distant clocks. Any independent tiora effect must be distinguished from the standard gravitational redshift/blueshift and Doppler shift already described by relativity.

Standard relativity already accounts for gravitational redshift, blueshift, and Doppler shift. A genuinely independent tiora prediction must therefore be a calculable residual shift or arrival-rate deviation that remains after mass and state of motion are specified.

Directional observation becomes model-discriminating only after the model can calculate the amplitude and parameter dependence of its deviation from GR.

28. Verified Theory, Open Problems, and the Domain of Tiora

GR and quantum theory have strong experimental support within their respective domains. The plausible research space for tiora lies mainly in regions that remain incompletely determined, including black-hole interiors, the terminal state of complete evaporation, mechanisms of information recovery, quantum gravity, the microscopic nature of dark energy, and possible strongly nonlinear temporal environments. This paper therefore separates verified results, extreme extrapolations, and open problems.

Research problem	**Status in existing physics**	**Candidate tiora explanation**	**Theoretical and observational requirement**
Weak-field and ordinary strong-gravity phenomena	GR has passed extensive precision tests; gravitational-wave and EHT results are highly consistent with Kerr/GR.	Treat GR as the macroscopic effective geometric description of tiora.	Recover GR in all measured regimes with deviations below observational limits.
Black-hole interior	The region inside an event horizon cannot be directly observed electromagnetically; classical GR contains singularities, while quantum effects are expected to become important.	Avoid an exact $q = 0$; the interior is an extremely slow but still evolving environment.	Provide a complete interior solution without arbitrary parameters.
Hawking evaporation	Predicted by semiclassical quantum field theory; complete evaporation of astrophysical black holes has not been directly observed.	Long external lifetime arises primarily from temporal output stretching and recovery.	Explain whether Hawking-like radiation still exists and what spectrum it has.
Information paradox	There is a deep unresolved tension between unitary quantum evolution and the picture of complete evaporation.	The slow-time branch assumes information remains present and enters the exterior causal domain at an extremely low rate.	Provide evolution consistent with quantum unitarity and energy conservation.
Event horizon	Observations strongly support Kerr-like exterior spacetime, but a “true horizon” is not an ordinary imageable surface.	An extremely low-q region can act as a long-lived causal-output bottleneck that approximates a horizon.	Satisfy constraints from shadow, ringdown, echoes, surface radiation, and related observables.
Nature of dark energy	Cosmic accelerated expansion is observed, but the microscopic nature of dark energy is unknown.	Tiora may be dynamical dark energy or an underlying manifestation of it.	Satisfy both cosmological data and local strong-gravity observations.
Direction of time-rate change and cross-region information mapping	Standard relativity does not use an absolute “cosmic time-rate field”; proper-time differences arise from motion and geometry.	Tiora uses the stable universe $q = 1$ as a reference and retains $q < 1$ and $q > 1$ after instability; the black-hole branch provisionally uses $q < 1$.	Derive the direction of q and the propagation factor P from coupling and propagation equations, while reproducing redshift, orbits, lensing, and clock experiments.

Table 1. Status of existing theory, candidate tiora explanatory paths, and the corresponding burden of proof. Open questions do not imply that GR or quantum theory as a whole lack experimental validation.

29. Explanatory Scope of the Current Model

At the present stage, tiora provides a unified candidate explanatory path rather than a mathematically closed or data-fitted theory. The q–Φ correspondence and q-based rewritings of Newtonian/GR weak-field formulas are consistency checks only; a full explanation requires tiora dynamics to derive these established relations independently. The present framework organizes the following phenomena into one causal chain:

- **Local consistency of proper time:** all local physical processes share the same τ, so a local observer does not directly feel an overall scaling of q.
- **Extreme time dilation:** when q = dτ/dt is extremely small, a finite local history maps to a very long external time.
- **Long apparent black-hole lifetime:** a short local recovery process may be unfolded by low q into a long external observational history.
- **Black-hole darkness:** the external arrival rate of new photons and other causal information approaches zero.
- **Continued existence of information:** as long as q remains positive, information continues to evolve locally and is output at an extremely low rate.
- **Recovery and apparent shrinkage:** R_physical decreases as the recovery front moves inward, while R_observed may lag for a long time because of delayed information.
- **Non-stellar formation paths:** a black-hole-like state depends on whether D crosses D_c and does not require a particular progenitor category.
- **Time-space coupling:** the spatial gradient of q and the deflection of worldlines jointly form the macroscopic description of geometry and gravity.

All of these remain components of an explanatory framework. Their scientific value depends on whether they can be developed into quantitative, falsifiable, cross-phenomenon predictions.

30. Falsifiable Predictions and Observational Tests

Tiora must correspond to explicit failure conditions. The current version can be constrained by the following observations and experiments:

1. The exterior geometry of black-hole candidates must be nearly identical to Kerr within current precision; otherwise EHT, stellar-orbit, and gravitational-wave data would exclude the model.
2. If the model adopts a branch without a strict event horizon, very-late-time signals may contain weak echoes, nonthermal output, or recovery-front effects absent from a standard Kerr black hole. If future observations constrain these signals below the minimum level permitted by the model, the corresponding tiora version is excluded.
3. The difference between R_physical and R_observed must produce a calculable time dependence consistent with multi-epoch observations.

4. If small-scale high-energy events can cross D_c from either direction in tiora space, they should generate extremely short anomalous delays, temporal compression, frequency shifts, or geometric responses. If high-energy experiments and astrophysical data constrain such effects below the model's minimum prediction, the model fails.
5. If tiora is identical to dark energy, local tiora disturbances in strong gravity must remain compatible with the large-scale uniformity of dark energy and with DESI, supernova, CMB, and related cosmological constraints.
6. The mass–tiora coupling equation and recovery equation must reproduce observations once parameters are fixed. If they cannot, the corresponding model version must be rejected.
7. The cross-region signal-playback rule must follow from one propagation equation. Once P_{B→A} is fixed, the same rule must explain observations in different directions and systems.
8. If an information catch-up effect exists, rapid tiora recovery should produce quantitatively predictable short-time arrival-rate compression, spectral changes, or energy-flux signatures. If observations exclude those signatures, the corresponding branch must be abandoned.

30.1 Observational Conditions That Would Increase the Credibility of Tiora

Compatibility with already known phenomena is only a necessary condition. Increased credibility requires tiora to predict, before observation, residual terms, threshold behavior, or cross-phenomenon relations not fixed by the standard theory, and for independent experiments or observations to confirm those predictions. Several particularly relevant classes of tests are:

1. **Reproducible deviations in high-precision clocks or high-energy systems consistent with the proposed energy-coupling mechanism.** If, after known SR/GR contributions and systematic errors are controlled, an additional time term remains that varies reproducibly with energy-conversion history, energy density, or energy-momentum state, it would directly test the weak-disturbance hypothesis of Section 20.3.
2. **Threshold-like temporal behavior near D_c.** For $D < D_c$ the deviation should remain small and smooth; after crossing D_c, time-arrival rate, frequency shift, or output rate should enter a distinctly nonlinear regime.
3. **Direct measurement of input synchronization–output dilation across q boundaries.** A single P_{B→A} should describe different experiments and propagation directions.
4. **A common additional q-delay function shared by multi-messenger signals.** GW170817 and its associated gamma-ray burst arrived only about 1.7 seconds apart after traveling more than one hundred million light-years, strongly constraining ordinary-universe differences between the speed of light and the speed of gravity.[23] Tiora must satisfy these constraints when $D \ll D_c$. Independent signals in extreme environments that share an additional delay governed by the same q(r,t) would have much greater discriminating power.
5. **Predictable very-late-time echoes from black-hole candidates.** Their delay, spectrum, and damping should be specified in advance by q, R_tear, and the recovery equation. Echoes alone are not unique, because other horizonless models and some GR treatments can also produce them.

6. **A white-hole-like extreme temporal-compression event.** Such an event should exhibit flux, spectral behavior, and instability timescales consistent with $q \gg 1$ or information catch-up. If it also satisfies the macroscopic causal characteristics of a GR white hole, the two mechanisms could be compared directly.
7. **A reproducible tiora-threshold signal in high-energy systems approaching the relativistic speed limit, or an effective superluminal phenomenon that still obeys an independent local q structure.** This would directly test the open branch coupling the light-speed barrier to the temporal environment.

The strongest support would come from one tiora equation set, with one frozen parameter set, explaining multiple independent phenomena. If each phenomenon requires an independent temporary rule, the claimed unification loses force.

31. Minimal Dynamical Sketch and Theoretical Completeness

As a minimal representation of recovery, let the local deviation D decrease in its own proper time τ according to a recovery function $\mathcal{R}$:

$$\partial \mathrm{D}/\partial \tau = -\mathcal{R}(\mathrm{D}, \nabla \mathrm{T}, \ldots)$$

In the slow-time black-hole branch, $d\tau = q\, dt$, so the recovery rate measured in external reference time automatically contains q:

$$\partial \mathrm{D}/\partial \mathrm{t} = -\mathrm{q}(\mathrm{D}) \cdot \mathcal{R}(\mathrm{D}, \nabla \mathrm{T}, \ldots)$$

This relation expresses the temporal mapping in the slow-time branch: the deeper D is and the smaller q becomes, the slower the same local recovery process appears in external reference time. If future equations support a $q > 1$ branch, the external-time mapping changes accordingly; the direction of $q(D, \mathrm{sign}\,\delta)$ must be determined by dynamics and data.

A spherically symmetric model also requires transport or diffusion terms depending on ∇T. A complete theory should use a conservation-law form and specify whether tiora carries energy, pressure, and a finite propagation speed. Parameters without physical justification are not fixed here.

31.1 Nature of the Present Equations: Definitions, Compatibility Rewrites, and Missing Dynamics

The mathematical relations currently used in this paper fall into three classes. First are **definitions**, including $q = d\tau/dt$, $\delta = (T - T_0)/T_0$, $D = |\delta|$, and the critical threshold D_c. Second are **compatibility expressions**, including $q \approx 1 + \Phi/c^2$, $\nabla^2(c^2 \ln q) \approx 4\pi G\rho$, and $a \approx -c^2 \nabla \ln q$ in the weak field; these are largely q-variable rewritings of known Newtonian/GR results. Third are the **tiora dynamics that do not yet exist**: how stress-energy changes T, how T and D determine q, the dynamical origin of D_c, cross-boundary propagation, and the recovery law for unstable regions.

The current version has not derived Newtonian gravity, SR, or Einstein's field equations from tiora first principles, nor has it demonstrated that mass, dark energy, or any known physical entity is identical to tiora. The present achievements are variable definitions, candidate causal structures, partial consistency checks against known physics, and a list of falsifiable tasks.

If tiora cannot independently derive established results, its role is limited to re-description. If it can derive established results but produces no distinguishable new predictions, it may still function as an interpretive or effective modeling tool. Only if it both recovers verified results and produces calculable predictions in open regimes that are fixed in advance and later confirmed would it qualify for serious consideration as a direction for extending relativity.

31.2 Necessary Work from Conceptual Framework to Testable Theory

For tiora to enter a serious theoretical-physics discussion, at minimum it must complete the following mathematical and experimental tasks:

1. **Specify the mathematical degrees of freedom of tiora.** It must be determined whether T is a scalar field, vector/tensor structure, effective-medium variable, or deeper geometric quantity. Its units, local observables, transformation properties, and stable state T_0 must be defined. If the model introduces a preferred reference structure, the associated Lorentz violation and experimental bounds must be stated. If local Lorentz symmetry is preserved, it must follow from the equations.
2. **Establish the source coupling between matter-energy and tiora.** A simple mass density ρ is insufficient to reproduce the source structure of GR. The model must at least treat energy density, momentum, pressure, and stress, and specify a coupling between the stress-energy tensor $T_{\mu\nu}$ and tiora. Candidate pictures in which mass is a source of tiora, a stable tiora deformation, or a manifestation of a common deeper structure must be selected by equations rather than preference.
3. **Establish tiora dynamics and conservation laws.** A closed evolution equation must determine propagation of T, diffusion and recovery of disturbances, propagation speed, possible energy and pressure carried by tiora, and conservation of total energy-momentum. Black-hole recovery, information catch-up, and white-hole-like high-rate output should emerge from the same dynamics.
4. **Derive SR independently from tiora.** If kinematic time dilation is interpreted as the macroscopic manifestation of a weak tiora disturbance, the model must derive the Lorentz factor $\gamma = 1/\sqrt{(1 - v^2/c^2)}$, frame symmetry, velocity addition, relativity of simultaneity, and proper-time differences between worldlines. It must also explain why the coupled state persists after acceleration ends without continuous power input.
5. **Derive the verified limits of GR independently from tiora.** The weak-field regime must recover the Newtonian inverse-square law, gravitational redshift, and Shapiro delay. Solar-System scales require perihelion precession and light deflection; rotating systems require frame dragging; dynamic strong fields require gravitational-wave propagation consistent with observation; and static or rotating black-hole exteriors must approach Schwarzschild/Kerr within present precision.
6. **Build strong-field equations and derive D_c.** The model must determine the dynamical origin of D_c, the nonlinear transition mechanism, the conditions selecting the slow-time and fast-time branches, and the explicit functional form $q(D, \operatorname{sign}\delta)$. Only then can “spacetime tear” become a quantitative prediction rather than a definition.

7. **Provide numerical predictions for black holes, echoes, white holes, and information propagation.** R_tear(t), R_observed(t), redshift, photon arrival rate, ringdown corrections, echo delays and damping, the spectrum and flux of information catch-up, and the stability timescale of white-hole-like output should all be calculated in advance and tested across multiple objects with the same parameter set.
8. **Establish consistency with quantum theory.** If the model addresses black-hole information preservation, Hawking-like radiation, or extreme temporal structures, it must specify quantum-state evolution, unitarity, coupling to quantum field theory, vacuum stability, superluminal causal structure, and relevant energy conditions.
9. **Pass existing experimental constraints.** Atomic clocks, GPS, particle accelerators, equivalence-principle tests, binary systems, the Solar System, GW170817, multi-messenger astronomy, EHT, LIGO/Virgo/KAGRA, cosmological-background data, and dark-energy measurements jointly constrain the parameter space. Parameters must be fixed and then tested across independent data sets.
10. **Specify exclusion criteria in advance.** A model version should be rejected if no parameter region can reproduce both SR/GR and black-hole observations; if D_c must be adjusted arbitrarily from one system to another; if predicted echoes, information catch-up, R_physical − R_observed, or white-hole-like outputs are already below experimentally permitted levels; or if the dark-energy-identification branch conflicts with cosmological constraints.

31.3 Validation Thresholds for Entering Theoretical Physics

The development of tiora can be divided into three stages.

Stage 1: Conceptual framework. This is the status of the present paper: variables are defined, a unified causal structure is proposed, consistency conditions with known physics are stated, and falsifiable directions are listed, but no closed dynamical system yet exists.

Stage 2: Effective theoretical tool or interpretive framework. A closed mathematical system is established, the main results of SR/GR are derived from the model's own equations in verified regimes, and observations are described with controlled parameters. If the resulting mathematics is fully equivalent to GR, the value of tiora would lie mainly in a new choice of variables, computational method, intuitive structure, or engineering interpretation.

Stage 3: New research direction or extension of relativity. In addition to recovering verified results, the theory gives clear, nontrivial, pre-registered predictions in regions not yet decided by GR, quantum field theory, or current cosmology; those predictions are then confirmed by independent experiment or observation, while the same equations simultaneously explain other data.

The next stage of work is therefore to compress the minimum assumptions into a closed equation set and to prioritize experiments with the greatest exclusion power. The scientific value of the framework will be determined jointly by mathematical closure, parameter identifiability, recovery of established limits, and independent validation of new predictions.

32. Time Engineering: Local Slowing, Relative Acceleration, and One-Way Travel to the Future

If future technology can manipulate tiora so that a local region is held stably in a controlled tear regime $D \geq D_c$ and the slow-time branch maintains $0 < q \ll 1$, then the observer's thought processes, bodily processes, clocks, and other local physical processes share the same proper time τ. Local experience remains normal, while less proper time is accumulated than in the $q = 1$ external universe. The corresponding relation is

$$\Delta t = \int d\tau/q.$$

Under an approximately constant low q, one local day could correspond to one external year, one century, or longer. After leaving the low-q region, the observer emerges at a later time in the external universe. This constitutes one-way travel to the future in the slow-time branch.

Different worldlines accumulating different proper times already occurs in SR and GR. The additional tiora time-engineering hypothesis is that this difference in time rate can be associated with an environmental variable that can be manipulated deliberately, so that future travel need not rely on near-light-speed flight or naturally occurring strong-gravity objects.

If tiora can be engineered stably, one-way future travel can be expressed simply as the controlled reduction of one's own time rate relative to the ordinary universe.

Time "acceleration" and "slowing" are also relative. If A is adjusted to $q = 0.01$ while B remains at $q = 1$, then measured on A's timescale, B advances 100 times faster; measured by B, A advances at 0.01 times the rate. Therefore, even if nature never permits a true $q > 1$ state, relative time acceleration can still be obtained by slowing the reference system.

The paper also retains $q > 1$ as an open branch. If some controllable tiora state allows $q > 1$, the local region would accumulate more than one second of proper time during one second of ordinary-universe reference time, constituting genuine local time acceleration. This branch currently lacks established dynamics.

Tiora time engineering therefore has at least three levels: first, generate local time slowing with $q < 1$; second, use relative comparison so that an unslowed region appears time-accelerated relative to a slow-time region; third, determine whether a true $q > 1$ local acceleration branch exists. The first level alone is sufficient for one-way travel to the future; the latter two require new propagation rules and experiments.

33. Conclusion and Research Program

This paper defines tiora as a speculative cosmic environmental variable associated with the stable existence and progression of time. Its ontology may correspond to a field, an effective environmental component, or a deeper rule, with no microscopic form imposed at present. The ordinary universe is assigned the stable reference state T_0, $q = 1$, and the highest environmental integrity.

Tiora values either above or below the stable range can increase the deviation magnitude D. When $D \geq D_c$ and q undergoes a nonlinear departure from 1, the state is defined as a spacetime tear. The black-hole model adopts the slow-time working branch in which q falls sharply, while retaining the q-increasing branch as an open possibility.

In the slow-time black-hole branch, a black hole is described as a severely unstable tiora region that continuously recovers from the outside inward. Matter, energy, and information continue to evolve locally as long as $q > 0$, while subsequent state updates to the external universe are stretched to extreme durations. Milliseconds or seconds of local history may correspond to an extremely long observational history for a distant universe. Darkness arises from the greatly reduced arrival rate of photons and other causal information, while $R_{observed}$ may lag $R_{physical}$ for long periods because of delayed historical information.

The framework has structural correspondences with horizonless black-hole mimickers, the black-hole information problem, the terminal stage of Hawking evaporation, short-interior/long-exterior timescales in quantum-gravity proposals, indications of dynamical dark energy, and black-hole formation paths that do not require ordinary stellar-remnant seeds. These correspondences establish context and constraints only; they are not independent evidence for tiora.

The theoretical status of tiora depends on mathematical closure and experimental testing. Any version that cannot simultaneously satisfy verified relativistic results and strong-gravity observations should be rejected. If a closed equation set can recover SR/GR in known regimes and make distinguishable predictions in open regimes that are later confirmed, the framework could develop from a conceptual model into an effective physical theory.

At the current stage, tiora has not derived SR, GR, or Newtonian gravity from first principles, and it lacks a complete mass-energy/tiora coupling equation, D_c dynamics, an explicit q(D) function, and calibrated recovery parameters. Weak-field relations such as $\nabla^2(c^2 \ln q) \approx 4\pi G\rho$ are variable rewritings of known gravitational formulas and serve only as consistency conditions. The central future task is to construct upstream dynamics that allow one parameter set to reproduce known physics while also generating independently testable new predictions.

If tiora is also engineerable, the slow-time branch could be used to create local low-q environments and produce one-way future travel. Relative time acceleration can be obtained by lowering q in a reference region, while genuine local acceleration with $q > 1$ remains an open branch.

Structurally, a uniform stable environment corresponds to SR-like local spacetime; mild gradients must reproduce GR/Newtonian free fall and orbits; extreme deviations crossing D_c enter the spacetime-tear regime. White-hole-like states are modeled as candidate high-q or information-catch-up output compression, whose high flux and stability must be calculated within the same dynamics.

The light-speed barrier is retained as a strong-field coupling problem. The value c remains the experimentally verified local causal limit of standard relativity. Tiora additionally asks whether extreme energy-momentum states near that limit can drive D toward D_c and whether a hypothetical effective superluminal phenomenon would appear through low-q output, spacetime tearing, or

causal-structure reconstruction. This branch acquires physical meaning only if equations fixed in advance gain experimental support.

Appendix A. Model Revision Path

This appendix records the major revisions from the earliest intuition to the current tiora model, clarifying changes in definition, abandoned branches, and the origins of the present assumptions.

Step 1: Time and change. The research began from the question of whether time must be defined or recognized through change, turning time from an assumed background into an object requiring further modeling.

Step 2: Searching for an environmental variable. Early versions borrowed the terms dark matter and dark energy. Because these terms already carry precise physical meanings, the model later adopted the independent name tiora.

Step 3: The black hole as an extreme temporal environment. An early version treated the black hole as a region in which time nearly stops and briefly associated $q = 0$ with matter and information losing existence.

Step 4: Information and conservation constraints. The deletion-style interpretation at $q = 0$ conflicts with quantum unitarity and conservation of energy/information, motivating a re-examination of whether q must ever reach zero.

Step 5: q only approaches zero. The slow-time branch was revised so that finite energy can only drive q continuously toward zero. Matter, energy, and information remain present while their external output becomes extremely slow.

Step 6: Recovery begins at formation. Recovery is assumed to begin as soon as the black-hole-like state forms and to proceed from the outside inward. Deep low-q regions recover more slowly in external reference time.

Step 7: Black-hole lifetime as temporal unfolding. A recovery history lasting only seconds locally may map to an extremely long apparent external lifetime under low q.

Step 8: Separation of physical state and observed state. The model introduced $R_{physical}$ and $R_{observed}$ to distinguish the recovery of the actual unstable region from delayed historical information reaching a distant observer.

Step 9: Recasting the information problem as an output-timescale problem. The focus shifted from whether information exists to the timescale on which it enters the external causal domain.

Step 10: Two-sided deviation. Both tiora above and below the stable interval are allowed to generate instability, making $D = |\delta|$ the threshold-control variable.

Step 11: Separating environmental integrity from time rate. A decrease in environmental integrity no longer fixes the direction of q; both $q < 1$ and $q > 1$ branches are retained.

Step 12: Slow-time black-hole working branch. The $q < 1$ branch is used to study darkness, redshift, long apparent lifetime, and the causal-output bottleneck, while remaining explicitly a working hypothesis.

Step 13: Cross-region information mapping. The propagation factor $P_{B\to A}$ was introduced to describe retiming of signals that cross between regions with different tiora states.

Step 14: Information catch-up. Whether rapid recovery from low q to $q = 1$ causes compressed replay of earlier output was isolated as a separate testable branch.

Step 15: Relative time engineering. Lowering q for A makes B appear time-accelerated relative to A, while true $q > 1$ remains a stronger open branch.

Step 16: Future travel. An engineerable stable low-q region directly yields a candidate mechanism for one-way future travel.

Step 17: Input synchronization–output dilation. Cross-region propagation was separated from the simple q_A/q_B ratio, with $P_{B\to A}$ representing possible receiver-side remapping.

Step 18: Unified limiting requirements for SR, GR, and ordinary gravity. Stable uniform tiora must recover SR; weak gradients must recover GR/Newtonian gravity; and the spatial gradient of q is used as a compatibility language for terrestrial gravity and planetary orbits.

Step 19: Airborne atomic clocks and energy coupling. The current version treats energy conversion and changes in energy-momentum state during acceleration as potential sources of tiora coupling and interprets SR kinematic time dilation as a weak-disturbance macroscopic limit. The branch still has to derive Lorentz symmetry and the γ factor independently.

Step 20: Light-speed barrier and threshold. The model retains the conjecture that extreme energy-momentum states may push D toward D_c and considers effective superluminal behavior associated with low-q output or causal-structure reconstruction.

Step 21: White-hole-like temporal compression. GR white holes retain their standard definition. Tiora proposes only a candidate underlying mechanism: high q or information catch-up may generate extreme outward output, a broadband near-white flash, and strong instability.

Step 22: Evidence standard. Cloud-9 motivated rejection of a simple identification between tiora and dark matter. Echoes, dynamical dark energy, and direct collapse are treated only as structural correspondences. Support for the model must come from results calculated in advance and governed by the same equation across independent phenomena.

Step 23: Status of weak-field formulas. The relations $\nabla^2(c^2 \ln q) \approx 4\pi G\rho$ and $a \approx -c^2\nabla \ln q$ can be obtained from existing Newtonian/GR weak-field relations and are therefore classified as consistency limits rather than new tiora laws.

Step 24: Theoretical completeness requirements. The model now explicitly lists the degrees of freedom, stress-energy coupling, conservative dynamics, independent derivation of SR/GR, strong-field D_c predictions, quantum consistency, and cross-data-set falsification required to move from a conceptual framework toward an effective theory.

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Reading Note

The main text consistently distinguishes existing observations, standard theoretical results, and tiora conjectures. EHT, LIGO, JWST, DESI, NASA, and other sources are used to establish empirical constraints, research context, or structural correspondence. None of these sources has detected or validated tiora.

The research program proceeds from the environmental hypothesis, stable interval, and critical deviation through the slow-time black-hole branch, cross-region information mapping, information catch-up, time engineering, and falsifiable predictions. The scientific significance of the model depends on whether these assumptions can be compressed into a closed equation set that allows observations to confirm or exclude clearly defined branches.